

Document No:	Title :Project Technical Appraisal	Number of Pages :
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1. PURPOSE

To assist in the technical appraisal of network reconfiguration, extension and reinforcement projects.

2. SCOPE

This procedure applies to all internal and external capital projects in ZETDC Eastern Region.

3. RESPONSIBILITIES

3.1 The Engineering Manager is responsible for ensuring that this procedure is implemented, maintained and regularly reviewed in line with the changes in operations, technology, legislation, standards, circumstances and policies.

4. DEFINITIONS

4.1 **GM** - General Manager

4.2 **EM**- Engineering Manager

5. ACTIVITY DESCRIPTION

- 5.1.1 The Engineer prepares conceptual designs for each proposed solution.
- 5.1.2 The Engineer shall derive estimate cost for each proposed solution.
- 5.2 The Engineer shall model and simulate the proposed solution in the power system simulation software.
- 5.3 For technical losses which are above 10%, the engineer suggests other solutions.
- 5.4 For technical losses which are below 10%, the engineer proceeds to investigate the voltage profiles
- 5.5 For voltages outside the 10% tolerance range, the engineer shall apply voltage control mechanisms and check the voltage again.
- 5.6 For voltages within the 10% tolerances, the engineer proceeds to check if capacity is available.
- 5.7 Where there is no capacity, the engineer suggests other solutions.
- 5.8 Where capacity is available, the engineer **proceeds does** fault level analysis.
- 5.9 For unacceptable fault levels, the engineer suggests other solutions.
- 5.10 For acceptable fault levels, the engineer then carries out the financial appraisal.

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6. RECORDS

The following records shall be generated from implementation of this procedure.

Name of record:	Custodian / location:	Retention period:
Project Appraisal Report	Engineer	5yrs
Network simulation case files	Planning Office	5yrs

After the retention period of 5 years, all the records shall be archived for a period of 30 years.

7. REFERENCES

- 7.1 Engineering Instructions
- 7.2 Distribution Planning Manual
- 7.3 Design Memo
- 7.4 Distribution Code
- 7.5 Project Appraisal Framework

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